DIABETES TECHNOLOGY & THERAPEUTICS
Volume 16, Number 5, 2014
© Mary Ann Liebert, Inc.
DOI: 10.1089/dia.2013.0222

DTT
Diabetes Technology & Therapeutics

ORIGINAL ARTICLE

Accuracy and Acceptability of the 6-Day Enlite Continuous Subcutaneous Glucose Sensor

Timothy S. Bailey, MD,[1] Andrew Ahmann, MD,[2] Ronald Brazg, MD,[3] Mark Christiansen, MD,[4] Satish Garg, MD,[5] Elaine Watkins, DO, MSPH,[6] John B. Welsh, MD, PhD,[7] and Scott W. Lee, MD[7]

Abstract

Objective: This study evaluated the performance and acceptability of the Enlite® glucose sensor (Medtronic MiniMed, Inc., Northridge, CA).
Subjects and Methods: Ninety adults with type 1 or type 2 diabetes wore two Enlite sensors on the abdomen and/or buttock for 6 days and calibrated them at different frequencies. On Days 1, 3, and 6, accuracy was evaluated by comparison of sensor glucose values with frequently sampled plasma glucose values collected over a 12-h period. Accuracy was assessed at different reference glucose concentrations and during times when absolute glucose concentration rates of change were <1, 1–2, and >2 mg/dL/min. The sensor's ability to detect hypoglycemia or hyperglycemia was evaluated with simulated alerts. Subject satisfaction was evaluated with a 7-point Likert-type questionnaire, with a score of 7 indicating strong agreement.
Results: With abdomen sensors under actual-use calibration (mean, 2.8 ± 0.9 times/day), the overall mean (median) absolute relative difference (ARD) values between sensor and reference values were 13.6% (10.1%); the corresponding buttock sensor ARD values were 15.5% (10.5%). With abdomen sensors under minimal calibration (mean, 1.2 ± 0.9 times/day), the mean (median) ARD values were 14.7% (10.8%). Mean ARD values of abdomen sensors at rates of change of <1, 1–2, and >2 mg/dL/min were 13.6%, 12.9%, and 16.3%, respectively. With abdomen sensors, 79.5% and 94.1% of hypoglycemic and hyperglycemic events, respectively, were correctly detected; 81.9% and 94.9% of hypoglycemic and hyperglycemic alerts, respectively, were confirmed. The failure rates for abdomen and buttock sensors were 19.7% and 13.9%, respectively. Mean responses to survey questions for all subjects related to comfort and ease of use were favorable.
Conclusions: The Enlite sensor provided accurate data at different glucose concentrations and rates of change. Subjects found the sensor comfortable and easy to use.

Introduction

The Enlite® continuous glucose monitoring (CGM) sensor (Medtronic MiniMed, Inc., Northridge, CA) was developed to enhance patient comfort and accuracy. The sensor is thin (31 gauge, with a 27-gauge introducer needle) and short (8.5 mm, with a 10.5-mm introducer needle), has a working electrode area of 0.39 mm^2, and has an implanted volume of 1.06 mm^3. The sensor and its introducer needle enter the skin at a 90° angle, after which the introducer needle is automatically retracted into a protective housing for convenience and safety. The sensor includes three electrodes

[1]AMCR Institute, Escondido, California.
[2]Oregon Health and Sciences University, Portland, Oregon.
[3]Rainier Clinical Research Center, Renton, Washington.
[4]Diablo Clinical Research, Walnut Creek, California.
[5]Barbara Davis Center for Childhood Diabetes, University of Colorado Denver, Aurora, Colorado.
[6]Profil Institute for Clinical Research, Chula Vista, California.
[7]Medtronic MiniMed, Northridge, California.

This study is registered at ClinicalTrials.gov with clinical trial registration number NCT01464346.
Portions of this work were presented at the 72nd Scientific Sessions of the American Diabetes Association, held in Philadelphia, Pennsylvania, June 8–12, 2012, the 21st Annual Scientific and Clinical Congress of the American Association of Clinical Endocrinologists, held in Philadelphia, Pennsylvania, May 23–27, 2012, and the 48th Annual Meeting of the European Association for the Study of Diabetes, held in Berlin, Germany, October 1–5, 2012.

plated on a flexible substrate, where a constant voltage is maintained across the working and reference electrodes. Glucose oxidase is spread across the electrodes, and the catalyzed reaction produces one electron per oxidized glucose molecule, resulting in a current that is proportional to the interstitial fluid glucose concentration.[1] Its design features were intended to provide an optimized glucose oxidase layer, good responsiveness at low oxygen levels, low dependence on temperature, good dynamic response, and resistance to the effects of interfering compounds. The high surface area of its sensing electrodes was intended to provide stability and linearity. We evaluated Enlite sensor performance over its anticipated working life of 6 days at different glucose concentrations, rates of glucose concentration change, and calibration frequencies.

Research Design and Methods

The study was conducted at seven U.S. investigational centers in 90 adult subjects 18–75 years of age with type 1 or type 2 diabetes. Inclusion criteria for subjects with type 1 diabetes included an age at onset of <40 years, exclusive use of insulin for diabetes management, and a history of normal weight or underweight at the time of diagnosis. Inclusion criteria for subjects with type 2 diabetes included an age at onset of ≥40 years, initial use of a glucose-lowering agent other than insulin, and a history of being at least overweight at the time of diagnosis. Exclusion criteria included an episode of diabetic ketoacidosis within 6 months of the screening visit, a history of a seizure disorder, cardiovascular disease, or adrenal insufficiency, or a hematocrit value lower than the normal reference range. Baseline hemoglobin A1C values were determined at a central laboratory by certified National Glycohemoglobin Standardization Program methodology.

The study included a run-in phase and a study phase. During the up-to-30-day run-in phase, subjects were trained in sensor insertion and removal, as well as in use of the additional study devices and study procedures. Subjects were supplied with Revel™ 2.0 pumps (Medtronic), which were not to be used to infuse insulin or manage the subjects' diabetes during the study; rather, the pumps were used to receive, store, and calibrate data from the sensors and blood glucose meters. Following the run-in phase, subjects were to participate in a 6-day study phase, for which they were assigned to wear two Enlite sensors on the abdomen, two on the buttock, or one in each area. Subjects were to be equally randomized among the three groups. Sensors were then to be connected to MiniLink™ transmitters (Medtronic) (integrated with the Revel 2.0 pumps). Subjects were instructed to continue with their current diabetes regimen independent of the study devices. Sensors were to be worn continuously for the duration of the trial.

The study phase included inpatient sessions on Days 1, 3, and 6. Each inpatient session was to last 12 h, during which the sensors were to be calibrated at different frequencies to determine the effect of calibration frequency on sensor accuracy. All calibrations were to be done with capillary whole blood glucose measurements from a Bayer Contour® NEXT LINK meter (Bayer HealthCare, Tarrytown, NY). During the outpatient intervals, subjects were to be asked to calibrate both sensors three or four times throughout the day to represent calibrations as they occur in routine use. During the inpatient visits, each subject was required to calibrate one sensor every 12 h and to calibrate the other sensor three or four times throughout the day. These calibration procedures were not subject to randomization; all subjects performed both types of calibration, and the two pumps used by each subject were identical. Calibrations were allowed during times of rapid glucose concentration change. Plasma samples were required to be obtained every 5–15 min as part of the frequent sample testing (FST) procedure.

The primary end point was sensor accuracy with the minimum calibration requirements (every 12 h after the second calibration), and the secondary end point was sensor accuracy with three or four calibrations throughout the day. Sensor accuracy was validated using a reference method (YSI 2300 STAT Plus™ glucose and lactate analyzer; YSI Life Sciences, Yellow Springs, OH). Sensor glucose values were paired with contemporaneous (within ± 5 min) YSI reference plasma glucose values for accuracy determinations. During each FST, subjects with an established insulin sensitivity ratio and insulin–carbohydrate ratio were to undergo a hypoglycemic challenge (glucose lowered to a target of 50–75 mg/dL for approximately 2 h, including 30 min between 50 and 60 mg/dL) and a hyperglycemic challenge (glucose raised to a target of 180–400 mg/dL for approximately 2 h, including 30 min between 350 and 400 mg/dL). Data from these intervals were stratified according to the rate of YSI glucose concentration change and used to estimate sensor accuracy of abdomen sensors during times of rapid, moderate, and slow rates of change.

Simulated predictive and threshold alert rate analyses were performed to assess the detection of hypoglycemic events (YSI ≤70 mg/dL) and hyperglycemic events (YSI ≥250 mg/dL). Simulated low and high CGM alert settings are provided in the tables. Device survival was determined by two methods: (1) site subject reports of sensor operation and (2) device downloads based on the sensor's first valid electrical signal and the last sensor glucose value. To obtain a more representative estimate of real-life sensor accuracy, the relative contribution of each glucose concentration-specific mean absolute relative difference (ARD) value from the FSTs was weighted according to the prevalence of glucose concentration values observed in a large dataset of 1,093 patient-years of blood glucose data (CareLink™ database; Medtronic).

Questionnaires were conducted that rated subjects' agreements with various statements on a 7-point scale ranging from 1 (''strongly disagree'') to 7 (''strongly agree''). The surveys were not validated or tested for reproducibility. Adverse event reports were collected throughout the run-in and study phases.

Results

The 90 enrolled subjects had a mean ± SD age of 44.4 ± 13.8 years (range, 18–71 years), 49 (54.4%) were male, 79 (87.8%) were white, 65 (72.2%) had type 1 diabetes, and 25 (27.8%) had type 2 diabetes. Of the subjects with type 2 diabetes, nine required insulin, and 16 did not. Thirty-four subjects (37.8%) were naive to CGM, and 44 (48.9%) were naive to insulin pump therapy. The mean ± SD body mass index was 28.8 ± 7.4 kg/m^2, the mean A1C value was

8.0 ± 1.5%, and the mean duration of diabetes was 19.8 ± 12.5 years. Twenty-nine subjects were randomized to wear both sensors in the abdomen, 32 were randomized to wear one in the abdomen and one in the buttock, and 29 were randomized to wear both sensors in the buttock.

Abdomen sensor performance

During the study, 38 abdomen sensors were replaced, and data from this total of 122 abdomen sensors worn by 61 subjects were included in the analysis; the 61 subjects included 29 who wore two abdomen sensors and 32 who wore one abdomen sensor. The observed calibration frequency for sensors that were to have been calibrated every 12 h was 1.2 ± 0.9 times/day, and that for sensors that were to have been calibrated three or four times per day was 2.8 ± 0.9 times/day. Accuracy data are presented in Tables 1 and 2. Table 1 gives mean and median biases and mean and median ARDs between Enlite and YSI glucose concentrations at different reference ranges for the two calibration frequencies. Compared with a calibration frequency of 1.2 ± 0.9 times/day, the data indicate that calibrating 2.8 ± 0.9 times/day will result in a 1.1% improvement in mean ARD (14.7% with 1.2 ± 0.9 times/day vs. 13.6% with 2.8 ± 0.9 times/day). Table 2 gives agreement rates (the percentage of CGM readings within specified percentages of paired reference values) at different reference ranges for the two different calibration frequencies.

During intervals of slow absolute glucose concentration change (0 to <1 mg/dL/min, 5,745 data points), the mean ± SD ARD of abdomen sensors was 13.6 ± 13.9% (median, 10.1%). During intervals of moderate absolute glucose concentration change (1 to <2 mg/dL/min, 1,282 data points), the mean ± SD ARD of abdomen sensors was 12.9 ± 11.8% (median, 9.6%). During intervals of rapid absolute glucose concentration change (≥2 mg/dL/min, 381 data points), the mean ± SD ARD of abdomen sensors was 16.3 ± 14.8% (median, 12.9%).

Sensor accuracy was determined on each of the 3 days in which an FST was conducted. For the Day 1 FST, the mean ± SD ARD was 15.9 ± 14.9% (median, 12.2%), for the Day 3 FST the mean ± SD ARD was 11.8 ± 10.9% (median, 9.1%), and for the Day 6 FST the mean ± SD ARD was 13.2 ± 14.4% (median, 9.6%).

Sensor and corresponding plasma glucose values (measured within ±30 min of one another) were analyzed to estimate the Enlite sensor's ability to correctly detect existing or predicted hypoglycemia and hyperglycemia. A summary of the simulations of alerts within ±30 min using either the threshold alert alone or the combined threshold and

Table 1. Continuous Glucose Monitoring Biases and Absolute Relative Differences Versus Reference YSI Values, Abdomen-Only Insertion Site, at Different Calibration Frequencies and YSI Glucose Ranges

YSI reference range (mg/dL), calibration frequency	*Number of paired points*	*Bias (mg/dL)*		*Absolute relative difference (%)*[a]	
		Mean	*Median*	*Mean*	*Median*
Overall					
3–4 times/day	7,415	2.05	0.93	13.63	10.10
Every 12 h	7,432	0.75	0.74	14.71	10.80
<40[a]					
3–4 times/day	3	28.18	31.65	28.18	31.65
Every 12 h	3	27.18	27.65	27.18	27.65
40–60[a]					
3–4 times/day	618	6.50	5.45	10.06	7.85
Every 12 h	515	6.56	4.75	11.07	9.15
61–80[a]					
3–4 times/day	1,419	4.81	3.80	11.43	9.05
Every 12 h	1,349	1.36	−0.15	12.04	9.90
81–180					
3–4 times/day	3,241	1.95	1.72	12.38	9.09
Every 12 h	3,158	2.14	3.00	13.95	10.45
181–300					
3–4 times/day	1,648	−2.79	−2.47	11.72	8.96
Every 12 h	1,895	−3.02	−1.03	12.62	8.54
301–350					
3–4 times/day	325	−6.32	−5.36	11.36	8.43
Every 12 h	346	−6.50	−2.08	12.27	6.97
351–400					
3–4 times/day	137	−11.99	−9.23	13.48	10.11
Every 12 h	132	−12.06	−7.98	13.68	8.29
>400					
3–4 times/day	24	−28.38	−26.78	28.38	26.78
Every 12 h	34	−28.08	−19.87	28.08	19.87

Continuous glucose monitoring readings are within 40–400 mg/dL.
[a]For the YSI reference range ≤80 mg/dL, the differences in mg/dL are included instead of percentage differences.

Table 2. Percentage Agreement of Continuous Glucose Monitoring–YSI Paired Points Within YSI Glucose Ranges, Abdomen-Only Insertion Site, at Different Calibration Frequencies

		Percentage of CGM within			
YSI reference range (mg/dL), calibration frequency	*Number of paired CGM–YSI*	*15/15% of YSI*	*20/20% of YSI*	*30/30% of YSI*	*40/40% of YSI*
Overall					
3–4 times/day	7,415	72.7	83.7	93.2	97.0
Every 12 h	7,432	68.1	79.0	90.2	96.3
<40[a]					
3–4 times/day	3	33.3	33.3	33.3	66.7
Every 12 h	3	33.3	33.3	66.7	100.0
40–60[a]					
3–4 times/day	618	79.6	91.1	96.1	97.9
Every 12 h	515	76.3	85.8	92.4	99.0
61–80[a]					
3–4 times/day	1,419	76.3	86.9	94.9	96.9
Every 12 h	1,349	70.5	82.5	92.8	98.8
81–180					
3–4 times/day	3,241	70.9	80.7	91.8	96.9
Every 12 h	3,158	64.9	76.5	89.0	95.9
181–300					
3–4 times/day	1,648	72.5	84.9	94.1	97.6
Every 12 h	1,895	68.9	78.9	90.2	95.4
301–350					
3–4 times/day	325	71.4	84.9	94.5	98.2
Every 12 h	346	72.3	80.6	91.0	94.2
351–400					
3–4 times/day	137	63.5	79.6	89.1	94.9
Every 12 h	132	72.0	79.5	87.9	93.2
>400					
3–4 times/day	24	33.3	41.7	54.2	66.7
Every 12 h	34	38.2	52.9	67.6	79.4

Continuous glucose monitoring (CGM) readings are within 40–400 mg/dL.
[a]For the YSI reference range ≤80 mg/dL, agreement was based on 15/20/30/40 mg/dL.

predictive alert set to a 30-min prediction horizon is shown in Table 3, using data from abdomen sensors that were calibrated either three or four times per day or every 12 h. Consensus error grid analysis results of FST data from abdomen sensors that were nominally calibrated at different frequencies are given in Table 4.

Accounting for all 122 abdomen sensors used in the study, 102 were expected to last to the end of the FST on Day 6. Of the 20 sensors that were not included in the survival analysis, seven were used to replace sensors not intended for 6-day use (such as a sensor used to provide data for a missed Day 1 or Day 3 FST), seven were removed because their companion sensors failed, and six experienced precalibration errors and were removed before any FSTs. After calibration, 82.3% of abdomen sensors included in the survival analysis (84 of 102) operated for more than 5 days (i.e., for between 120 and 144 h). The mean functional life of abdomen sensors (excluding those that were removed because the companion sensor was removed) was 133.4 h, and their median functional life was 132.4 h.

During the FSTs, 23.5% of the reference values were in the hypoglycemic range, resulting in an overall mean ARD of 13.6%. The ''real-life'' percentages of time spent in each glucose concentration range, seen in the CareLink database, were 8.1% for ≤75 mg/dL, 59.1% for >75–180 mg/dL, and 32.8% for >180 mg/dL. When observed values were adjusted for this glycemic distribution, a projected overall mean ARD of 12.8% was calculated.

Buttock sensor performance

Buttock sensor data were analyzed separately. After calibration, 89.6% of buttock sensors operated more than 5 days. For buttock sensors nominally calibrated three or four times per day, the overall mean ± SD ARD was 15.5 ± 16.0%, and the overall median ARD was 10.5%. For buttock sensors nominally calibrated at 12-h intervals, the overall mean ± SD ARD was 18.2 ± 23.8%, and the overall median ARD was 12.2%. A comparison of the difference in accuracy for sensors inserted in the abdomen with that of sensors inserted in the buttock indicated that this difference was not statistically significant ($P = 0.28$); however, this result should be interpreted with caution as the study was not specially powered to test for this difference.

Overall safety for abdomen and buttock sensors

Adverse events related to the study device or procedures were mild, and there were no serious adverse events or unanticipated adverse device effects with the use of the Enlite sensor during the study. There was one study device-

Table 3. Detection and Prediction of YSI-Verified Hypoglycemic and Hyperglycemic Events Occurring Within ±30 min by Abdomen Enlite Sensors

	Threshold alerts				Threshold and predictive alerts			
	Calibrating 3–4 times/day		*Calibrating every 12 h*		*Calibrating 3–4 times/day*		*Calibrating every 12 h*	
Alert setting	*Events correctly detected (%)*	*Alerts verified by events (%)*	*Events correctly detected (%)*	*Alerts verified by events (%)*	*Events correctly detected (%)*	*Alerts verified by events (%)*	*Events correctly detected (%)*	*Alerts verified by events (%)*
≤60 mg/dL	63.1	67.5	70.2	51.4	89.0	50.3	86.3	39.7
≤70 mg/dL	79.5	81.9	83.1	74.5	93.2	67.0	92.5	61.8
≤80 mg/dL	91.0	85.4	89.8	83.5	97.2	73.5	96.5	72.2
≤90 mg/dL	95.4	89.3	94.9	87.6	98.8	78.7	97.3	76.9
≤100 mg/dL	96.3	91.6	95.4	87.8	98.9	82.5	98.1	78.8
≥180 mg/dL	94.1	94.9	91.1	94.6	97.5	88.3	94.6	88.6
≥220 mg/dL	92.4	91.5	90.1	92.8	95.1	84.5	94.3	85.2
≥250 mg/dL	87.4	93.2	87.9	92.3	94.6	85.5	94.5	84.5
≥300 mg/dL	77.0	87.5	82.0	84.7	88.5	73.5	89.1	73.2

Table 4. Consensus Error Grid Analysis of Abdomen Sensors

	Calibration frequency	
Consensus error grid	*3–4 times/day*	*Every 12 h*
Number of paired points	7,388	7,395
Zone A	86.3%	81.3%
Zone B	12.5%	17.3%
Zone C	1.1%	1.3%
Zone D	0.0%	0.1%
Zone E	0.0%	0.0%

related adverse event (pain at the sensor insertion site that resolved within a few hours of removal) and no unanticipated adverse device effects. Of the 432 sensors used (202 in the run-in phase and 230 in the study phase), 426 skin assessments were conducted following the removal of the sensor. In both the abdomen and buttock the majority of skin findings were due to redness from the sensor insertion, device, and/or adhesive. All skin findings were mild in nature, and none of the skin findings resulted in an adverse event. There were no unexpected lesions or skin irritation associated with sensor wear.

Overall satisfaction for abdomen and buttock sensors

Subjects' experiences with the Enlite sensor were generally favorable. On a 7-point Likert-type survey, with 7 indicating strong agreement with a particular statement, median scores of 6 or 7 were seen with each of the following statements: ''The sensor was comfortable under my skin,'' ''I did not feel the sensor underneath my skin,'' ''The sensor was easy to remove from my skin,'' ''The sensor started up reliably,'' ''The first day sensor performance was acceptable,'' ''I do not fear inserting these CGM sensors,'' ''The sensor insertion device was easy to use,'' ''Sensor insertion was pain-free,'' ''I like that the retractable needle protects me from injury,'' and ''I would recommend this CGM to other patients with diabetes.'' Only three statements elicited median scores of 4 (neutral), and none of the statements had a median response score of less than 4. Statements with a median score of 4 included ''I could insert the sensor in my buttock myself, without help from others,'' ''I like that I do not have to see the needle,'' and ''The hole in my skin caused by the sensor will heal more quickly than the hole from other sensors I've worn.''

Discussion

The Enlite sensor, when placed in the abdomen and calibrated three or four times per day, provides accurate readings over a 6-day interval in adults with either type 1 or type 2 diabetes, with an overall mean ARD of 13.6%. Less frequent calibration yielded results that were not markedly different than results obtained with more frequent calibration. Whereas prior generations of sensors required stable glucose concentrations for calibration, the present data were obtained with calibrations performed at any time, including when glucose concentrations were changing rapidly and when many glucose concentrations were in the high or low ranges. Enlite sensor glucose readings remained accurate at slow, moderate, and rapid rates of glucose concentration change.

The Enlite sensor's mean ARD was lower when placed in the abdomen than when placed in the buttock.

The study was designed to assess sensor accuracy at the beginning, middle, and end of sensor life. In order to collect adequate sensor data in the low and high glycemic ranges, subjects were artificially induced to have prolonged periods of time in the hypoglycemic and hyperglycemic ranges.

The study has several limitations. The assignment of each sensor to one of two possible calibration regimens was not subject to randomization, so subjects and/or investigators may have made the assignment in a way that biased the results. The study was not specifically designed or powered to compare the accuracy of abdomen sensors with the accuracy of buttock sensors. Additional subanalysis of sensor accuracy according to the calibration frequency (calibrated three or four times per day versus every 12 h) within different glucose ranges was presented descriptively only because of the small sample size within each grouping provided in Table 1.

CGM sensors are less accurate with extreme glucose concentrations approaching 40 or 400 mg/dL. By the design of the in-clinic experiments, this study obtained more low and high glucose values than are typically experienced. Therefore, the accuracy that patients experience in real life may be better. In order to approximate this, our data were normalized to those seen in the CareLink database. This resulted in an expected improvement in accuracy with a mean ARD of 12.8%. This report also demonstrates that the combination of threshold and predictive alerts, versus threshold-only alerts, results in improved detection of both hypoglycemic and hyperglycemic excursions. A recent study by Zijlstra et al.[2] evaluated 2,317 paired data points generated by the Medtronic Guardian REAL-Time CGMS and noted that the system failed to detect more than half of the true hypoglycemic events (sensitivity, 37.5%) and that more than half of the alarms that warn patients for hypoglycemia were false (false alert rate, 53.3%). With the Enlite sensor, the alarm performance is significantly improved, with 50.3% of combined threshold and predictive alerts for hypoglycemia $\leq$60 mg/dL being verified by hypoglycemic events within $\pm$30 min (Table 3). Further improvements in the percentage of events correctly detected are seen when a predictive algorithm is added. Patients using this sensor can configure the threshold and predictive alarms in a way that balances the benefits of being notified of out-of-range glucose values with the potential nuisance of a false alarm.

Previous studies have evaluated Enlite sensor performance characteristics. Keenan et al.[1] reported a mean ARD of 13.86%, with 97.3% of points within the A + B zones of the Clarke error grid. Calhoun et al.[3] reported higher ARD values than are reported here (18% and 15% for mean and median ARD values, respectively) and also observed that Enlite sensors under-reported glucose concentrations across the entire concentration range, whereas in the present study, the Enlite sensor bias was positive in the hypoglycemic range and negative in the hyperglycemic range. The reasons for this discordancy are unclear, although Calhoun et al.[3] analyzed data from eight inpatient studies of subjects with type 1 diabetes, some of whom were as young as 5 years of age; glucose data were restricted to nighttime measurements (10:00 p.m. to 6:00 a.m.), and the studies were not designed to primarily evaluate Enlite sensor accuracy. Luijf et al.[4] evaluated the accuracy of Enlite sensors in a clinical research center and at home in 20 patients with type 1 diabetes and found that in the clinical research center, the mean ARD was 16.4%, whereas at home, the mean ARD was 18.9%. The authors noted an insufficient amount of data pairs in the hypoglycemic area and suggest that more aggressive insulin dosing might be needed in future investigations.

Survey responses indicated that the Enlite sensor was easy to insert, easy to use, and comfortable to wear. Some patients had difficulty with unassisted buttock insertion, as indicated by the survey responses, and buttock sensor accuracy was generally less than abdomen sensor accuracy, suggesting the abdomen as the preferred site for Enlite wear. Observations such as this are important, as difficulties with sensor wear adherence with the Sof-Sensor® (Medtronic) have been reported.[5] Patients consider these factors as well as sensor accuracy when they plan their therapy.[6]

In conclusion, the Enlite sensor, when placed in the abdomen, provides accurate data over 6 days of use. Its accuracy is maintained at various glucose concentrations, glucose concentration rates of change, and calibration frequencies. It is durable, comfortable, safe, and easy to use. These performance characteristics, combined with algorithms to interrupt insulin delivery in response to hypoglycemia,[7] may assist in the pathway towards commercialization of closed-loop artificial pancreas systems.

Acknowledgments

The authors thank Francine R. Kaufman, MD, an employee of Medtronic, for comments on an earlier draft of this manuscript, and the subjects who participated in the study. This study was funded by Medtronic MiniMed, Inc., Northridge, CA.

Author Disclosure Statement

T.S.B., A.A., R.B., M.C., S.G., and E.W. have received research support from Medtronic, Inc. T.S.B. has received research support from Abbott, ACON, Alere, Animas, BD, Dexcom, Insulet, Lifescan, Lilly, and Tandem and has received consulting honoraria from Bayer, BD, and Medtronic, Inc. J.B.W. and S.W.L. are employees of Medtronic, Inc.

References

1. Keenan DB, Mastrototaro JJ, Zisser H, Cooper KA, Raghavendhar G, Lee SW, Yusi J, Bailey TS, Brazg RL, Shah RV: Accuracy of the Enlite 6-day glucose sensor with Guardian and Veo calibration algorithms. Diabetes Technol Ther 2012;14:225–231.
2. Zijlstra E, Heise T, Nosek L, Heinemann L, Heckermann S: Continuous glucose monitoring: quality of hypoglycaemia detection. Diabetes Obes Metab 2013;15:130–135.
3. Calhoun P, Lum J, Beck RW, Kollman C: Performance comparison of the Medtronic Sof-Sensor and Enlite glucose sensors in inpatient studies of individuals with type 1 diabetes. Diabetes Technol Ther 2013;15:758–761.
4. Luijf YM, Mader JK, Doll W, Pieber T, Farret A, Place J, Renard E, Bruttomesso D, Filippi A, Avogaro A, Arnolds S, Benesch C, Heinemann L, DeVries JH; AP@Home

Consortium: Accuracy and reliability of continuous glucose monitoring systems: a head-to-head comparison. Diabetes Technol Ther 2013;15:722–727.

5. Slover RH, Welsh JB, Criego A, Weinzimer SA, Willi SM, Wood MA, Tamborlane WV: Effectiveness of sensor-augmented pump therapy in children and adolescents with type 1 diabetes in the STAR 3 Study. Pediatr Diabetes 2012;13:6–11.
6. Ramchandani N, Arya S, Ten S, Bhandari S: Real-life utilization of real-time continuous glucose monitoring: the complete picture. J Diabetes Sci Technol 2011;5:860–870.
7. Bergenstal RM, Klonoff DC, Garg SK, Bode BW, Meredith M, Slover RH, Ahmann AJ, Welsh JB, Lee SW, Kaufman FR; ASPIRE In-Home Study Group: Threshold-based insulin-pump interruption for reduction of hypoglycemia. N Engl J Med 2013;369:224–232.

Address correspondence to:
John B. Welsh, MD, PhD
Medtronic MiniMed, Inc.
18000 Devonshire Street
Northridge, CA 91325

E-mail: john.b.welsh@medtronic.com